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Attachments to the Public Switched Telephone Network (PSTN)

Category II attachment requirements for 1200 bits per second duplex modems standardized for use on the PSTN

(The text of this ETS may be utilized, wholly or in part, for the establishment of NET 22)

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Foreword

This European Telecommunication Standard (ETS) has been produced by the Terminal Equipment (TE) Technical Committee of the European Telecommunications Standards Institute (ETSI). The text of this ETS may be utilized, wholly or in part, for the establishment of NET 22.

This ETS contains the technical requirements for approval to Category II of 1200 bits per second duplex modems standardized for use on the Public Switched Telephone Network (PSTN). Approval and testing to Category II is applicable only at the request of the applicant. These requirements are based on, and do not conflict with, CCITT Recommendation V.22 [1].

Additional requirements are included which relate to end-to-end inter-operability over PSTN connections. These requirements are in excess of the CCITT Recommendations. A modem which complies with CCITT Recommendation V.22 [1] should always meet the requirements of this ETS which relate to parameters specified in that CCITT Recommendation.

Clause 4 of this ETS references the requirements which are common to both Category I and Category II modems.

Clause 5 of this ETS contains Category II requirements specific to 1200 bits per second duplex modems. In the case of certain functions common to a number of different types of modem (e.g. Auto-answering sequence) reference is made to Clause 5 of ETS 300 114 [2] which contains the relevant requirements.

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1 Scope

This ETS specifies the technical characteristics to be met by modems seeking Category II approval for duplex operation over the PSTN at 1200 bits per second. The modulation scheme specified is that described in CCITT Recommendation V.22 [1].

NOTE: CCITT Recommendation V.22 [1] also describes a method of transmitting data at 600 bits per second. This ETS specifies the characteristics to be met by such modems and, if requested by the applicant, this feature can also be approved as Category II.

The term "modem" in the context of this ETS includes all physical implementation practices for voice band modems which are galvanically connected to the PSTN.

This ETS specifies four modes of operation each with five modes of use (see subclause 5.2).

This ETS also contains descriptions of the tests to be performed in order to confirm compliance with the functional requirements contained herein. A general description of the test conditions and test requirements is given in Annex A (Normative).

2 Normative references

This ETS incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to, or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] CCITT Recommendation V.22 (1984): "1200 bits per second duplex modem standardized for use in the General Switched Telephone Network and on point-to-point 2-wire leased telephone-type circuits".

[2] ETS 300 114 (1990): "Attachments to the Public Switched Telephone Network (PSTN); Basic attachment requirements for modems standardized for use on the PSTN".

[3] Draft prETS 300 001 (1990): "Attachments to the Public Switched Telephone Network (PSTN) ; General technical requirements for equipment connected to an analogue subscriber interface in the PSTN".

[4] CCITT Recommendation V.25 (1984): "Automatic answering equipment and/or parallel automatic calling equipment on the general switched telephone network including procedures for disabling of echo control devices for both manually and automatically established calls".

[5] CCITT Recommendation S.33 (1984): "Standardization of an international text for the measurement of the margin of start-stop machines using International Alphabet No 5".

[6] CCITT Recommendation V.52 (1984): "Characteristics of distortion and error-rate measuring apparatus for data transmission".

3 Definitions and abbreviations

The abbreviations and definitions of ETS 300 114 [2] apply, together with the following.

3.1 Definitions and abbreviations

Answer mode: when calls are established with automatic facilities, a standard answer mode shall be used by the modem at the answering station. This mode consists of conventional characteristics (e.g. use of high channel carrier frequency or particular scrambler generating polynomial) complementary to those used in the standard call mode by the modem at the calling station, in order to ensure proper connection and interworking.

If calls are established on the PSTN by operators, or for leased line operation, bilateral agreement on the use of call mode and answer mode shall be necessary.

Call mode: when calls are established with automatic facilities, a standard call mode shall be used by the modem at the calling station. This mode consists of conventional characteristics (e.g. use of low channel carrier frequency or particular scrambler generating polynomial) complementary to those used in the standard answer mode by the modem at the answering station, in order to ensure proper connection and interworking.

If calls are established on the PSTN by operators, or for leased line operation, bilateral agreement on the use of call mode and answer mode shall be necessary.

Data Terminal Equipment (DTE): the expression "DTE" used to define the origin and destination of signals present at the digital interface of a modem. This expression does not require that a "commercial data terminal" be present to receive or generate such signals; a tester or any other suitable device may monitor or generate such signals.

Modem: a functional unit that modulates and de-modulates signals in order to enable digital data to be transmitted over analogue transmission facilities.

Modem conformance tester (MCT): essentially a modem to the same recommendation as the modem under test, but the individual sub-systems within it are both accessible (e.g. provide test points and permit functions to be enabled or disabled when required) and externally controllable (e.g. permit sequences such as the start up procedure to be selectively repeated). The sub-systems within a conformance tester may be constructed as discrete items of equipment, so as to permit their assembly into varying configurations required to suit the tests (e.g. the asynchronous to synchronous converter may be simply applied to a synchronous CCITT Recommendation V.22 [1] conformance tester to achieve an asynchronous V.22 conformance tester).

As an interim measure, until the conformance tester is defined, its definition agreed to be appropriate by ETSI, and such a tester is available, a modem used for reference may be used in its place. In the absence of previous approval to Category II of the modem used for reference, in the relevant modes of use/operation, the testing authority shall ensure that the modem used for reference complies with the relevant ETS to the extent necessary for the performance of the test.

Modem used for reference: a modem used for some of the tests specified herein or in another modem specific ETS. A modem used for reference may, at the discretion of the applicant, be provided by the testing authority or by himself. It shall be designed:

- to meet the requirements of the same Recommendation(s) of the CCITT as the modem under test, to the extent necessary for performing the tests;

- to provide the functionalities for a modem used for reference that are specified in the relevant testing Clauses; and

- to provide an interface which is accessible and of a type suitable for use in the tests (e.g. CCITT Recommendation V.24).

Where the applicant has provided the modem used for reference and the test fails, the testing authority may not be in a position to determine the precise reason for failure.

Modes of operation: are defined as modes specified in this modem specific ETS, that have an influence upon line signals present at the PSTN interface.

Modes of use: are specified in this modem specific ETS, that have an influence upon conditions present at a digital interface e.g. a "conventional" CCITT Recommendation V.24 interface or a PC bus interface in the case of an integral modem .

On-line state: the on-line state or condition is defined as an electrical condition into which, when connected to the network, a modem is placed such that it draws enough current to be capable of activating the exchange.

NOTE: Usually, a modem in the on-line state is potentially capable of sending or receiving speech-band information to or from the network.

Silence: in the context of this ETS, silence is defined as signals which in the relevant frequency band have an in-band power level which is at least 30 dB below the level of the transmitted signal at the point of measurement. This term is used to describe periods where signals are not transmitted during the hand-shaking sequences.

3.2 Abbreviations

AMM	Answer Mode Modem
Cct	Circuit
CEPT	Conférence Européenne des Administrations des Postes et des Télécommunications
CMM	Call Mode Modem
DCE	Data Circuit-Terminating Equipment
DTE	Data Terminal Equipment
MCT	Modem Conformance Tester
PSTN	Public Switched Telephone Network

4 General requirements

4.1 References to other ETSs

The modem shall comply with ETS 300 114 [2], Clause 4.

NOTE: ETS 300 114 [2] in turn refers to Draft prETS 300 001 [3] for the majority, if not all, of its requirements.

4.2 Information to be provided by the applicant

4.2.1 Information required for testing purposes

The applicant shall declare for which of the modes of operation/use identified in this ETS approval to Category II is sought.

Compliance shall be considered to have been accomplished by provision of the relevant information.

NOTE: This could be accomplished by completion of forms such as those provided in Annex C.

4.2.2 Instructions for use

Instructions for use shall be made available with the apparatus. The instructions for use shall include :

a) the apparatus or types of apparatus to which the instructions apply;

b) any information specifically indicated in this ETS for inclusion in the "Instructions for use"; and

c) any national restrictions on the use of the apparatus.

Any additional information that has been included shall be disregarded unless it is the subject of another ETS.

Compliance shall be checked by inspection.

5 Functional requirements specific to category II modems

5.1 General requirements

To comply with the requirements of this ETS, the modem shall provide:

- duplex mode of operation with continuous carrier, by means of the modulation scheme specified in CCITT Recommendation V.22 [1];

- channel separation by frequency division;

- means by which the channels may be selected either manually and/or automatically, where a modem is capable of transmitting in either of the two channels;

- provision of a guard tone of 1800 Hz ± 20 Hz, which a modem is to transmit while transmitting in the high channel;

- signalling rates of 1200 bit/s and, optionally, 600 bit/s.

NOTE 1: The general requirements described above are a subset of CCITT Recommendation V.22 [1]. In the requirements which follow any deviations from the strict interpretation of the Recommendation have been identified.

NOTE 2: In the following, references are made to interchange circuits between the modem and the Data Terminal Equipment (DTE), as defined in CCITT Recommendation V.24. However, not all modems provide an interface with such interchange circuits. For these cases the references to CCITT Recommendation V.24 type interchange circuits indicate equivalent functionality of a DTE and a modem.

NOTE 3: To perform certain tests specified herein, it is desirable that it be possible to disable the transmit scrambler of the modem.

5.2 Modes

It shall be possible to configure the modem to operate in at least one of the following modes of operation:

- 1200 bit/s call mode;

- 1200 bit/s answer mode;

and, optionally, in one or both of the following modes of operation:

- 600 bit/s call mode;

- 600 bit/s answer mode.

For each of the modes of operation at least one of the following modes of use shall be provided:

a) asynchronous with 8 bits per character;

b) asynchronous with 9 bits per character;

c) asynchronous with 10 bits per character;

d) asynchronous with 11 bits per character;

e) synchronous.

NOTE: It is acceptable for the same mode of use to be provided for each of the modes of operation selected.

In the asynchronous (start-stop) modes of operation, the modem shall accept a data stream from the DTE at a nominal rate of 1200 bits per second or, optionally, 600 bits per second. The asynchronous data to be transmitted shall, in accordance with subclause 5.9, be converted into a synchronous data stream suitable for transmission.

Demodulated data shall be decoded in accordance with subclause 5.5, then descrambled in accordance with Clause 5.10 and it shall then be passed for re-conversion into an asynchronous data stream in accordance with subclause 5.9.

In the synchronous mode of operation, the modem shall accept synchronous data from the DTE. The data shall then be scrambled in accordance with subclause 5.10 and passed to the modulator for encoding in accordance with subclause 5.5.

Demodulated data shall be decoded in accordance with subclause 5.5, then descrambled in accordance with subclause 5.10.

It shall be the applicant's responsibility to specify for which of these modes of operation/use approval for Category II is sought.

5.3 Line signals

5.3.1 Carrier frequencies

The carrier frequencies shall be 1200 Hz ± 0,5 Hz for the low channel and 2400 Hz ± 1 Hz for the high channel.

Compliance shall be checked by the method described in Annex A, Clause A.2.

5.3.2 Guard tone

5.3.2.1 Guard tone - call mode modem

When the modem is transmitting in the low channel, it shall not transmit the guard tone specified in Clause 5.3.2.2.

Compliance shall be checked by the method described in Annex A, Clause A.3.1.

5.3.2.2 Guard tone - answer mode modem

When the modem is transmitting in the high channel, a guard tone at 1800 Hz ± 20 Hz shall be transmitted simultaneously. The level of the guard tone shall be 6 dB ± 1 dB below the level of the data signal in the high channel.

Optionally, facilities may be provided to disable the transmission of the guard tone of 1800 Hz. Since the provision of the guard tone is mandatory, all tests in this ETS shall be carried out with guard tone present.

Compliance shall be checked by the method described in Annex A, Clause A.3.2.

The overall level of the guard tone and the data signal of a CCITT Recommendation V.22 [1] modem shall remain in compliance with the in-band spectral power limits for the network to which they are being connected (See ETS 300 114 [2], Clause 4).

NOTE: The purpose of the guard tone is to inhibit the false operation of in-band signalling systems such as in international networks (see CCITT Recommendation Q.144). It is thought by some that a marginal performance advantage may be gained as a result of the slightly higher transmit level possible when the guard tone is disabled and therefore this option is permitted.

5.3.3 Transmitted spectrum

The national network specific spectral power limits are specified in ETS 300 114 [2], Clause 4.

NOTE: ETS 300 114 [2], Clause 4 in turn refers to Draft prETS 300 001 [3]. There are no requirements in this ETS relating to the spectrum of signals presented to the PSTN over and above those for PSTN access invoked by subclause 4.1.

5.4 Line signalling rates

The signalling rates transmitted to line shall be 1200 bit/s ± 0,01% and, if requested by the applicant for approval to Category II, 600 bit/s ± 0,01%.

Compliance shall be checked by the method described in Annex A, Clause A.4.

5.5 Encoding and decoding of data

The encoding and decoding of data shall be in accordance with CCITT Recommendation V.22 [1], Section 2.5.2 and Table 1/V.22.

Until a Modem Conformance Tester (MCT) is available, compliance shall be checked by inspection of the applicant's declaration of conformance. When an MCT is available, compliance shall be checked by the method described in Annex A, Clause A.5.

5.6 Threshold of received line signal detection

The modem shall comply with the requirements of ETS 300 114 [2], subclause 5.4, when receiving a data signal corresponding to scrambled binary 1 at 1200 bit/s in the respective receive channel for the modem.

Compliance shall be checked by the method described in Annex A, Clause A.6.

5.7 Channel allocation

A Call Mode Modem (CMM) shall be capable of transmitting signals in the low channel and responding to signals received in the high channel. An Answer Mode Modem (AMM) shall be capable of transmitting signals in the high channel and responding to signals received in the low channel.

5.7.1 Channel selection

A modem which is capable of being configured as both a CMM and an AMM shall provide facilities for at least one of the following techniques of channel selection:

a) manual selection of the channels using facilities provided on the modem;

b) selection of the channels by the DTE (equivalent: Cct 126 control);

c) automatic selection of the channels as described in subclause 5.7.2.

NOTE: Other means of channel selection may also be provided but verification of the correct functioning of such facilities is not a requirement of this ETS.

For techniques a) and b), compliance shall be checked by the method described in Annex A, Clause A.7.1.

5.7.2 Automatic channel selection

A modem providing this facility shall, in the absence of manual intervention or a contrary command via a digital interface (e.g. an equivalent to control using Cct 126), be automatically configured as:

a) a CMM when exchanging data on-line to another compatible modem, and when the modem has entered the on-line state other than in response to an incoming PSTN call;

b) an AMM when exchanging data on-line to another compatible modem, and when the modem has entered the on-line state in response to an incoming PSTN call.

Compliance shall be checked by the method described in Annex A, subclause A.7.2.

5.8 Hand-shaking sequences

In the following sequences the signal described shall be sent continuously unless the description indicates that a period of silence be present.

5.8.1 Auto-calling and auto-answering procedures

5.8.1.1 Auto-calling - calling tone

This ETS does not require the provision of the calling tone defined in CCITT Recommendation V.25 [4].

5.8.1.2 Auto-calling - recognition of answering tone

For modems to be used for automatically originated calls, the modem, or its associated call establishment equipment, shall comply with the requirements for answering tone detection given in ETS 300 114 [2], subclause 5.2.1.

5.8.1.3 Auto-answering

For modems to be used for automatically answered calls, the modem, or its associated call answering equipment, shall comply with the requirements for answering tone generation given in ETS 300 114 [2], subclause 5.2.2.

5.8.2 Start-up procedures

Initially, the modem shall be indicating that

a) it is not detecting a valid carrier signal (equivalent: Cct 109 OFF),

b) it is not ready to transmit data (equivalent: Cct 106 OFF)

NOTE: For a modem to indicate that it is ready to transmit data (equivalent: turning ON Cct 106) requires that the DTE has indicated that it wishes to transmit data (equivalent: turning ON Cct 105).

5.8.2.1 Call mode modem

After connection to line, and when presented with unscrambled binary 1 and the guard tone from the remote modem (AMM), the CMM shall remain silent for a minimum of 551 ms. The modem shall then transmit scrambled binary 1.

NOTE: The value of 551 ms is derived from CCITT Recommendation V.22 [1] and is not the same as the value for CCITT Recommendation V.22bis. (see Annex B).

When a change from unscrambled binary 1 to scrambled binary 1 occurs in the data received by the CMM, it shall continue to transmit the scrambled binary 1 for a further period of 1035 ms ± 50 ms.

At the end of this period, the CMM shall

a) give an indication to the DTE that the modem has detected a valid carrier signal (equivalent: turning ON Cct 109);

b) give an indication to the DTE that the modem is now ready to transmit data (equivalent: turning ON Cct 106);

c) enter the data transmission phase.

For a CMM, condition a) should always occur before condition b) although the criteria for test is simply that these indications are now ON.

Compliance shall be checked by the method described in Annex A, subclause A.8.1.

5.8.2.2 Answer mode modem

On completion of the CCITT Recommendation V.25 [4] auto-answering sequence or, when manually answered, on connection to line, the AMM shall transmit unscrambled binary 1 at 1200 bit/s and the guard tone. When a change from silence to scrambled binary 1 at 1200 bit/s occurs in the line signal received by the AMM, it shall continue to transmit unscrambled binary 1 at 1200 bit/s and the guard tone for a further 270 ms ± 40 ms followed by scrambled binary 1 at 1200 bit/s and the guard tone for 765 ms ± 10 ms.

At the end of this period, the AMM shall:

a) give an indication to the DTE that the modem has detected a valid carrier signal (equivalent: turning ON Cct 109);

b) give an indication to the DTE that the modem is now ready to transmit data (equivalent: turning ON Cct 106);

c) enter the data transmission phase.

For an AMM, conditions a) and b) should occur almost simultaneously. The order is therefore of no importance.

Compliance shall be checked by the method described in Annex A, subclause A.8.2.

5.9 Transmission of start-stop characters

If the modem is capable of transmitting start-stop characters without error correction, speed conversion or flow control, it shall provide at least one of the following modes of use at 1200 bit/s and, where requested by the applicant for approval to Category II, at 600 bit/s:

a) asynchronous with 8 bits per character;

b) asynchronous with 9 bits per character;

c) asynchronous with 10 bits per character;

d) asynchronous with 11 bits per character.

The asynchronous-to-synchronous conversion shall be in accordance with ETS 300 114 [2], subclause 5.5.

NOTE: The tests for compliance with the above requirements are described in ETS 300 114 [2], Annex B, Clause B.5.

5.10 Scrambler and descrambler

A self synchronizing scrambler and a self synchronizing descrambler as specified in CCITT Recommendation V.22 [1], subclauses 5.1 and 5.2 shall be provided in the transmitting part and the receiving part, respectively, of the modem.

This implies that:

- the scrambler shall include circuitry to detect a sequence of 64 consecutive binary 1s at its output and, if detected, invert the next bit input to the scrambler;

- the descrambler may or may not include a circuitry to detect sequences of 64 consecutive binary 1s at its input and, when detected, to invert the next bit output from the descrambler. Where such a circuitry is included, it shall not begin operating prior to the initial hand-shaking sequence being completed. Furthermore, where this circuitry is included, the detection of the test loop 2 initiating signal of unscrambled binary 1s (see subclause 5.11.1, b)) shall be performed before the inversion described above can take place.

Compliance shall be checked by the method described in Annex A, subclause A.9.

5.11 Test loop 2

The modem shall provide facilities for the implementation and the remote control of test loop 2. The description of test loop 2 may be found in ETS 300 114 [2], subclause 5.3.2.

NOTE: The inter-DCE signalling for control of test loop 2 as specified in CCITT Recommendation V.54 has not been adopted for modems according to Recommendation CCITT V.22 [1]. Instead a different procedure has been defined in that Recommendation.

In the following, the sequences for the remote control of test loop 2 are defined irrespective of the line signalling rate (600 bit/s or 1200 bit/s).

5.11.1 Instigation of a remote loop 2

a) controlling modem

The modem which is instructed, either manually or by the DTE (equivalent: turning ON Cct 140), to instigate a remote loop 2, shall transmit an initiation signal of unscrambled binary 1.

When the modem is presented with scrambled reversals it shall detect them and, after not less than 231 ms, and not more than 308 ms transmit scrambled binary 1.

When presented with scrambled binary 1, the modem shall indicate to the user or to the DTE (equivalent: turning ON Cct 142) that the modem is in a test mode.

Compliance shall be checked by the method described in Annex A, subclause A.10.1.1

b) controlled modem

When the modem is presented with the initiation signal of unscrambled binary 1, transmitted by the remote modem, for not less than 154 ms and not more than 231 ms, it shall detect them and transmit scrambled reversals.

When presented with scrambled binary 1, the modem shall activate loop 2 and indicate to the user or to the DTE (equivalent: turning ON Cct 142) that the modem is in a test mode.

Compliance shall be checked by the method described in Annex A, subclause A.10.1.2.

5.11.2 Termination of a remote loop 2

a) controlling modem

When the modem from which a remote loop 2 had been instigated is instructed to terminate that loop (equivalent: turning OFF Cct 140, where the remote loop 2 instigation was controlled by the DTE), it shall suppress the line signal for 77 ms ± 10 ms after which data transmission shall be restored. The modem shall indicate to the user or to DTE (equivalent: turning OFF Cct 142) that the modem is no longer in a test mode .

Compliance shall be checked by the method described in Annex A, subclause A.10.2.1

b) controlled modem

When the modem inside which a loop 2 was remotely instigated is presented with a loss of line signal for 17 ms ± 7 ms followed by the reappearance of the signal, it shall de-activate loop 2 and restore normal operation within 172 ms ± 57 ms, after which data transmission shall be restored. The modem shall indicate to the user or to DTE (equivalent: turning OFF Cct 142) that the modem is no longer in a test mode.

Compliance shall be checked by the method described in Annex A, subclause A.10.2.2

5.12 Receiver performance

When tested as described in ETS 300 114 [2], subclause 5.6, the modem shall accumulate not less than 75% error free seconds during each of the performance testing periods.

NOTE 1: Since CCITT Recommendation V.22 [1] does not define performance criteria, the requirements of this clause are additional to those of the CCITT Recommendation.

NOTE 2: The figure of 75% is a provisional value, and based on reasoned estimations.

Annex A (normative): Testing methods

A.1 General testing conditions

A.1.1 General notes

The general conditions for test shall apply, as described in ETS 300 114 [2], Annex B.

For the testing of the modem it shall generally be necessary to simulate the PSTN in the test set-up in order for the modem under test and the modem conformance tester to remain in an on-line state. Unless otherwise indicated, Test Line 3 (see ETS 300 114 [2], Annex C, Clause C.5) shall be used to connect the modem under test and the modem conformance tester.

The following information is given to aid the testing authority in determining the signals present on the line.

Unscrambled binary 1 at 1200 bit/s is characterised by tones at 1050 Hz and 1650 Hz for the low channel; and 2250 Hz and 2850 Hz for the high channel.

Whatever the binary input, scrambled binary signals are characterised by a general distribution of signal power throughout the frequency band allocated to the particular mode (i.e. 650 Hz to 1750 Hz for call mode and 1850 Hz to 2950 Hz for answer mode). Therefore the binary content of the transmitted data signal cannot be determined without demodulation.

Where it is required to confirm that the data transfer phase has been established a single message shall be transmitted in each direction. There shall be no need to specifically check the text or data for errors since all Category II modems are subjected to a performance check. The test message shall be:

- for asynchronous modems, " THE QUICK BROWN FOX...." text in the international alphabet N° 5 (IA5), according to CCITT Recommendation S.33 [5];

  NOTE 1: Alternatively, the French version of the test text (VOYEZ LE BRICK GEANT...) as specified in CCITT Recommendation S.33 [5] may be used.

  NOTE 2: Either the 64-character set version or the 95-character set version of the test texts as specified in CCITT Recommendation S.33 [5] may be used.

- for synchronous modems, approx 1022 bits of pseudo-random data.

A.1.2 Proposed order for performing the tests

Except where otherwise stated, tests may be carried out in any order. The order of tests shall be at the discretion of the testing authority.

The following proposal should therefore only be taken as a guideline:

NOTE: The following presentation is used for the designation of entries summarized hereinafter:

Keyword / Clause / Subclause where the requirement is specified / Clause where the test is specified (or other means of conformance check).

1 Carrier frequencies / subclause 5.3.1 / Clause A.2;

2 Encoding and decoding of data / subclause 5.5 / Applicant's declaration of conformance;

3 Auto answering sequence / subclause 5.8.1 / [ETS 300 114 [2], subclauses 5.2.1 and 5.2.2];

4 Start-up procedures / subclauses 5.8.2.1 and 5.8.2.2 / subclauses A.8.1 and A.8.2;

5 Transmitted spectrum / subclause 5.3.3 / ETS 300 114 [2], Clause 4 ; Draft prETS 300 001 [3];

6 Channel allocation / subclauses 5.7.1 and 5.7.2 / subclauses A.7.1 and A.7.2;

7 Line signalling rates / subclause 5.4 / Clause A.4;

8 Guard tone, call mode modem / subclause 5.3.2.1 / subclause A.3.1;

9 Guard tone, answer mode modem / subclause 5.3.2.2 / subclause A.3.2;

10 Threshold of received line signal detector / [subclause 5.6; ETS 300 114 [2], subclause 5.4] / Clause A.6;

11 Scrambler and descrambler / subclause 5.10 / Clause A.9;

12 Asynchronous to synchronous converter / subclause 5.9 / [ETS 300 114 [2], Clause B.5];

13 Instigation to a remote test loop 2 / subclause 5.11.1 a) / subclause A.10.1.1;

14 Termination of a remote test loop 2 / subclause 5.11.2 a) / subclause A.10.2.1;

15 Response to an activate test loop 2 command / subclause 5.11.1 b) / subclause A.10.1.2;

16 Response to a terminate test loop 2 command / subclause 5.11.2 b) / subclause A.10.2.2;

17 General requirements / subclause 5.1 / Results of the relevant tests specified above;

18 Modes / subclause 5.2 / Results of the relevant tests specified above;

19 Receiver performance / subclause 5.12 / [ETS 300 114 [2], subclauses B.6.4.2 and B.6.4.3].

A.1.3 Limitation of number of tests

A.1.3.1 Introduction

Four modes of operation are described, each of which may have up to five modes of use.

The requirements as stated in this ETS are valid for all 20 possible modes. However, it is clear that the performance of all the tests in all the possible modes is both unnecessary and undesirable.

The following subclauses of A.1.3 define the combinations of tests which shall be performed in order to test conformity with this ETS.

In the remaining subclauses of A.1.3, the words "modes of operation" and "modes of use" refer only to those modes for which the applicant has requested Category II approval.

A.1.3.2 General rules

For each mode of operation, only one mode of use shall be tested. Where available, this shall be the synchronous mode. Where tests are performed in the asynchronous mode one character length shall be chosen using the following rule: the first choice is 10 bits/character, then 11 bits/character, then 9 bits/character and finally 8 bits/character.

A.1.3.3 Specific Rules

After selection using the general rules given in subclause A.1.3.2, the following specific rules shall be applied:

a) the following tests shall be performed, where relevant, for all modes of operation:

- carrier frequencies (requirement subclause 5.3.1 - test Clause A.2);

- guard tone (requirement subclause 5.3.2 - test Clause A.3);

- performance (requirement subclause 5.12).

b) The following tests shall be performed in both the call and answer modes of operation, where relevant, at the highest available data signalling rate for the mode being tested:

- auto-call and auto-answer (requirement subclause 5.8.1 - test Clause A.8);

- channel allocation (requirement subclause 5.7 - test Clause A.7);

- threshold of received line signal detector (subclause 5.6 - test Clause A.6).

c) The following tests shall be performed for one 1200 bit/s mode of operation and, where relevant, for one 600 bit/s mode of operation. Where the tests are to be performed at differing data signalling rates, where possible one test shall be performed in the call mode of operation and the other in the answer mode of operation:

- data signalling rates (requirement subclause 5.4 - test Clause A.4);

- encoding and decoding (requirement subclause 5.5 - test Clause A.5);

d) The following tests shall be performed for one 1200 bit/s mode of operation:

- transmission of start/stop characters (requirement subclause 5.9);

- scrambler and descrambler (requirement subclause 5.10 - test Clause A.9);

- test loop 2 (requirement subclause 5.11 - test Clause A.10).

A.2 Test for subclause 5.3.1 (carrier frequencies)

The following test may be performed in conjunction with the tests described in Clause A.7.

Tests A.2.2, A.2.3 and A.2.4 are specified in order to provide an alternative route for approval to Category II while awaiting the approval of an MCT for modems to CCITT Recommendation V.22 [1]. Once the MCT is available, tests should only be performed to subclause A.2.1

A.2.1 Test for subclause 5.3.1 using the MCT

The modem conformance tester and the modem under test shall be caused to enter the data phase. The modem conformance tester shall derive the carrier frequency from the received data signal. This can be done by demodulating the incoming signal with an accurately known carrier frequency. The frequency and phase of the carrier is then adapted until proper demodulation occurs. The measurement accuracy shall be 0 ± 0,1 Hz.

A.2.2 Test for modems where the scrambler can be inhibited

For a modem that is capable of operating as both a CMM and an AMM, these tests are performed in conjunction with the tests for subclause 5.7.1. The modem is caused to assume the appropriate mode of operation by one of the methods described in subclause A.7.1.

NOTE: Several attempts may be required before the desired result can be obtained because there is a 50% chance that the modem may detect the input '01' signal as '10' which is not the same.

For a modem that is capable of operating as a CMM, the modem is caused to transmit a signal of contiguous dibits of the form '01' (not '10') with the scrambler de-activate. The frequency transmitted by the modem to line shall be 1200 Hz ± 0,5 Hz.

For a modem that is capable of operating as an AMM, the modem is caused to transmit a signal of contiguous dibits of the form '01' (not '10') with the scrambler de-activate. The frequency transmitted by the modem to line shall be 2400 Hz ± 1,0 Hz.

A.2.3 Tests for modems where test points are accessible

If it is not possible to de-activate the scrambler in the modem under test, then the applicant shall designate points of measurement where a signal is accessible from which the carrier frequencies can be deduced.

A.2.4 Test for modems where neither A.2.2 nor A.2.3 applies

In cases where neither the scrambler can be inhibited nor test points with the property described in subclause above are available, the applicant shall submit a declaration of conformance to the testing authority stating that the requirements of subclause 5.3.1 are met.

A.3 Test for subclause 5.3.2 (guard tone)

The following test may be performed in conjunction with the test of Clause A.4.

A.3.1 Test for subclause 5.3.2.1 (guard tone, call mode modem)

The modem under test shall be configured as a CMM, the modem conformance tester shall configured as an AMM.

The PSTN simulator shall be conditioned to provide both modems with appropriate dial tone and DC conditions. An outgoing call attempt shall made from the modem under test to the modem conformance tester. With both modems in the on-line state and transferring data, the spectrum of the signal transmitted by the modem under test at its line terminals shall be checked to verify that no discrete spectral component measured in a 10 Hz bandwidth and emitted in the band 1780 Hz to 1820 Hz is greater than -32 dBm.

A.3.2 Test for subclause 5.3.2.2 (guard tone, answer mode modem)

The modem under test shall be configured as an AMM, the modem conformance tester shall be configured as a CMM. The PSTN simulator shall be conditioned to provide modems with appropriate ringing signals and DC conditions. Using the modem conformance tester or any other appropriate means, a call shall be established between the two modems. Once the modem under test is on line, and following, if present, the answering tone according to CCITT Recommendation V.25 [4], the spectrum of the signal transmitted to line is checked to verify that the guard tone as specified in subclause 5.3.2 shall be emitted in the band 1780 Hz to 1820 Hz, and that the level of the guard tone measured in a 10 Hz bandwidth is 6 dB ± 1 dB lower than the data signals transmitted in the band 1850 Hz to 2950 Hz. The level of the data signals shall be measured in a 1 kHz bandwidth, the centre frequency of which shall be 2400 Hz.

A.4 Test for subclause 5.4 (line signalling rates)

The following test may be performed subsequently to the tests described in subclause A.8. The modem is required to have completed the sequences described in subclause 5.8 before it reaches the state in which the requirements according to subclause 5.4 can be tested.

Call Mode Modem (CMM)

The modem conformance tester shall be configured to operate as an AMM in the synchronous mode of use.

After completion of the tests described in Clause A.8 for a modem under test which is a CMM, the modem under test is caused to enter the data transfer phase and transmit data.

The frequency of the signal appearing on the receiver element timing circuit (equivalent: Cct 115) of the modem conformance tester shall be measured. The average value of the frequency, measured over five periods of ten seconds, shall be within the limits of the line signalling rate specified in subclause 5.4.

Answer Mode Modem (AMM)

The modem conformance tester shall be configured to operate as a CMM in the synchronous mode of use.

After completion of the tests described in Clause A.8 for a modem under test which is an AMM, the modem under test shall enter the data transfer phase and transmit data.

The frequency of the signal appearing on the receiver element timing circuit (equivalent: Cct 115) of the modem conformance tester shall be measured. The average value of the frequency, measured over five periods of ten seconds, shall be within the limits of the line signalling rate specified in subclause 5.4.

A.5 Test for subclause 5.5 (encoding and decoding of data)

The Modem Conformance Tester and the Modem Under Test shall be caused to enter the data phase. A test pattern consisting of continuous binary 1 shall be transmitted for 125 s ± 5 s from the Modem Conformance Tester to the Modem Under Test. No errors shall occur during transmission of these bits. If errors occur, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test.

A.6 Test for subclause 5.6 (threshold of received line signal detection)

The test shall be performed as described in ETS 300 114 [2], Annex B, Clause B.4. The modem under test shall consecutively be configured to operate as an AMM and as a CMM (if applicable).

The signal to be applied to the modem under test is a data signal corresponding to scrambled binary 1 at 1200 bit/s in the respective receive channel for that modem.

A.7 Test for subclause 5.7 (channel allocation)

A suitable mechanism to assist discrimination between the signals transmitted by the modem conformance tester and the modem under test would be to arrange for the signals transmitted by the modem conformance tester to be at a lower level (e.g. 10 dB - 15 dB lower) than those of the modem under test.

For the purpose of the tests in this subclause a signal shall be present when total signal power within the specified band exceeds a level of -30 dBm.

For each of the tests in this subclause the modem under test shall be connected to the modem conformance tester via Test Line 3. Calls are established between the modem under test and the modem conformance tester and in each case the modem conformance tester shall be caused to assume the complementary mode to the modem under test.

The following tests shall be performed for each of the modes of operation which the applicant has identified:

A.7.1 Test for subclause 5.7.1 (general)

- for modems providing technique of channel selection (a), using the means of manual selection, the modem is caused to transmit a signal as a CMM. The spectrum of the signals transmitted by the modem under test shall be checked to verify that it contains signal power within the permitted range of 650 Hz to 1750 Hz, but not in the range 1850 Hz to 2950 Hz. The modem is now caused to transmit a signal as an AMM. The spectrum of the signals transmitted by the modem under test shall be checked to verify that it contains signal power within the permitted range of 1850 Hz to 2950 Hz, but not in the range 650 Hz to 1750 Hz.

- for modems providing technique of channel selection (b), using the information supplied by the applicant, the modem is caused to transmit a signal as a CMM, the spectrum of the signals transmitted by the modem under test shall be checked to verify that it contains signal power within the permitted range of 650 Hz to 1750 Hz, but not in the range 1850 Hz to 2950 Hz. The modem is now caused to transmit a signal as an AMM, the spectrum of the signals transmitted by the modem under test shall be checked to verify that it contains signal power within the permitted range of 1850 Hz to 2950 Hz, but not in the range 650 Hz to 1750 Hz.

- for modems providing technique of channel selection c), compliance is checked by the method described in the tests for subclause 5.7.2 (see subclause A.7.2)

During any one of the above tests, the checks of the frequencies required by the test of Clause A.2 may be performed.

A.7.2 Test for subclause 5.7.2 (automatic channel selection)

Call Mode Modem (CMM)

The modem conformance tester is configured to operate as an AMM in the synchronous mode of use. An outgoing call is made from the modem under test to the modem conformance tester. With both modems in the on-line state and transferring data, the spectrum of the signal transmitted by the modem under test is checked at the line terminals of the modem under test to verify that it contains signal power within the permitted range of 650 Hz to 1750 Hz, but not in the range 1850 Hz to 2950 Hz.

Answer Mode Modem (AMM)

The modem conformance tester is configured to operate as a CMM in the synchronous mode of use. The PSTN simulator is conditioned to provide the modem under test with appropriate ringing signals and, when the call is answered, to supply the DC conditions to the modem under test.

Once the modem under test is on line, and following, if present, the answering tone according to CCITT Recommendation V.25 [4], and both modems transferring data, the spectrum of the signal transmitted by the modem under test is checked at the line terminals of the modem under test to verify that it contains signal power within the permitted range of 1850 Hz to 2950 Hz, but not in the range 650 Hz to 1750 Hz.

A.8 Test for subclause 5.8 (hand-shaking sequences)

A.8.1 Call mode modem (CMM)

The modem conformance tester is configured to operate as an AMM in the synchronous mode of use.

A call is originated from the modem under test to the modem conformance tester. The modem conformance tester answers the call and generates the CCITT Recommendation V.25 [4] answer sequence. A data pattern other than continuous binary 1 is applied to the transmit data connection(s) of the digital interface (equivalent: Cct 103) of the modem under test.

The modem conformance tester then transmits unscrambled binary 1 at the relevant signalling rate accompanied by guard tone.

For not less than 551 ms and not more than 671 ms after the moment at which the modem conformance tester commences transmission of unscrambled binary 1, the total power of signals transmitted by the modem under test in the band 700 Hz to 1700 Hz shall be less than -50 dBm.

The modem under test shall then commence to transmit scrambled binary 1.

270 ms after the modem under test commences transmitting scrambled binary 1, the modem conformance tester ceases transmitting unscrambled binary 1 and commences transmitting scrambled binary 1 and the 1800 Hz guard tone.

The modem under test shall continue to transmit scrambled binary 1 for not less than 985 ms and not more than 1085 ms, measured from the moment at which the modem conformance tester commenced transmission of scrambled binary 1.

At the end of this period, the modem under test shall:

a) where applicable, complete any transaction with the DTE, which may be under-way in the dialogue mode;

b) give an indication to the DTE that a valid carrier signal has been detected (equivalent: turning ON Cct 109);

c) with the DTE signalling that it wishes to transmit data (equivalent: Cct 105 being ON), give an indication to the DTE that the modem is now ready to transmit data;

d) commence transmission of the data pattern being presented to its digital interface.

A.8.2 Answer mode modem (AMM)

The modem conformance tester is configured to operate as a CMM in the synchronous mode of use.

A call attempt is made from the modem conformance tester to the modem under test. Once the modem under test is on line, and following, if present, the answering tone according to CCITT Recommendation V.25 [4].

The modem under test shall transmit unscrambled binary 1 and the 1800 Hz guard tone.

The modem conformance tester is caused to transmit scrambled binary 1 at the relevant data signalling rate.

The modem under test shall, on receipt of the scrambled binary 1 and the 1800 Hz guard tone, continue to transmit unscrambled binary 1 for a further period of not less than 230 ms and not more than 310 ms.

The modem under test shall terminate the period specified above by transmitting scrambled binary 1 for not less than 755 ms and not more than 775 ms.

At the end of this period, the modem under test shall:

a) where applicable, complete any transaction with the DTE, which may be underway in the dialogue mode;

b) give an indication to the DTE that a valid carrier signal has been detected (equivalent: turning ON Cct 109);

c) with the DTE signalling that it wishes to transmit data (equivalent: Cct 105 being ON), give an indication to the DTE that the modem is now ready to transmit data;

d) commence transmission of the data pattern being presented to its digital interface.

A.9 Test for subclause 5.10 (scrambler and descrambler)

The tests described hereinafter shall be carried out at a data signalling rate of 1200 bit/s. If a modem which has not been approved as Category II is being used for reference, then it shall be possible to inhibit the scrambler and the descrambler in this modem, and to apply an external scrambler and descrambler, or a scrambler and descrambler integrated into the data test equipment. The scrambler and the descrambler external to the modem conformance tester shall have the properties specified in CCITT Recommendation V.22 [1], Section 5.

If the modem under test can be operated, for the purpose of test, in the synchronous mode of use, then it shall be configured to this mode.

With the scrambler and the descrambler enabled in the modem conformance tester, a call shall be initiated from the modem under test across Test Line 3 (see ETS 300 114 [2], Annex C, Clause C.5). After completion of the initial handshake sequence between the two modems and both modems being in the on-line state, the scrambler and the descrambler in the modem conformance tester shall be inhibited.

A.9.1 Test for modems which can be operated in the synchronous mode of use

The following tests shall be performed when the modem under test can be operated in the synchronous mode of use:

a) a constant binary 1 condition shall be transmitted from the modem under test to the modem conformance tester, and vice versa, for a duration of 125 s ± 5 s each (this may be done simultaneously if possible with the data test equipment used). The data received and descrambled at either end of the connection shall be verified for the appearance of binary 0 conditions. No more than three bits with a binary 0 value shall be received at either end. If more bits with a binary 0 value are received, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test;

b) the test of a) shall be repeated with a constant binary 0 condition. The data received and descrambled at either end of the connection shall be verified for the appearance of binary 1 conditions. No more than three bits with a binary 1 value shall be received at either end. If more bits with a binary 1 value are received, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test.

c) a 511-bit data test pattern according to CCITT Recommendation V.52 [6] shall be transmitted for a period of 125 s ± 5 s from the modem under test to the modem conformance tester (and through the external descrambler, if present), and vice versa (through the external scrambler, if present). No more than three bits error in either direction shall be encountered under this test. If more than three bit errors is encountered, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test.

A.9.2 Test for modems which can only be operated in a asynchronous mode of use

These tests shall only be performed if the modem under test cannot be operated in the synchronous mode of use, for the purposes of test.

NOTE: The tests specified hereinafter may be performed in conjunction with the tests of the asynchronous to synchronous converter as specified in ETS 300 114 [2], Annex B, subclauses B.4.3. If, however, the test fails, it is not clear whether the cause of the failure was the scrambler/descrambler or the asynchronous to synchronous converter. For this reason, where possible, it is preferable to test the scrambler and the descrambler in the synchronous mode of use.

An asynchronous to synchronous and synchronous to asynchronous converter shall be inserted between the scrambler and descrambler used with the modem conformance tester and the data source and sink. The data test equipment shall be capable of recognizing the standard test text (see ETS 300 114 [2], Annex B, subclause B.4.1.2) and the discrete start-stop characters (DTCs) N° 1 and N° 3

(see ETS 300 114 [2], Annex B, subclause B.4.1.3), and indicating whether sequences of characters are correctly received.

a) DTCs N° 3 shall be transmitted from the modem under test to the modem conformance tester, and vice versa for a period of 125 s ± 5 s. The characters received at either end of the connection shall be verified for their correctness. No more than one character shall be received corrupted at either end. If more than one corrupted character has been received, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test.

b) The test a) shall be repeated with DTCs N° 1.

c) The standard test text shall be transmitted from the modem under test to the modem conformance tester and vice versa for a period of 125 s ± 5 s. Not more than one character of the test text shall be received corrupted at either end. If more than one corrupted character has been received, the test shall be repeated once. If any errors occur during the second attempt, the modem shall be assumed to have failed the test.

A.10 Test for subclause 5.11 (test loop 2)

A.10.1 Instigation of remote test loop 2

A.10.1.1 Transmission of the test loop initiation signal (controlling modem under test)

The modem under test shall be in the on-line state and transmitting data signals at a data signalling rate of 1200 bit/s (e.g. after performing the tests of specified in Clause A.2.). Using the means described by the applicant, the modem is commanded to issue a request for remote test loop 2. The line signals transmitted and received by the modem under test are monitored to observe the following:

- as a result of the command the modem under test shall change the transmitted signal from scrambled data to unscrambled binary 1;

- the modem conformance tester is caused to respond by sending contiguous scrambled reversals;

- within a time of not less than 231 ms and not more than 308 ms, starting from the reception of a line signal equivalent to scrambled reversals to its line terminals, the modem under test shall transmit scrambled binary 1.

- upon receipt of its own scrambled binary 1 from the modem conformance tester, indicating the activation of the remote loop, and for not less than 231 ms and not more than 308 ms, the modem under test shall now indicate to the terminal that the modem is in a test mode (equivalent: turning ON Cct 142).

A.10.1.2 Response to the test loop initiation signal (controlled modem under test)

The modem under test is conditioned to be in the on-line state and to transmit data signals (e.g. after performing the tests according to Clause A.2). The line signals transmitted and received by the modem under test shall be monitored to observe the following:

- upon receipt of unscrambled binary 1 for not less than 154 ms and not more than 231 ms from the modem conformance tester, the modem under test shall transmit contiguous scrambled reversals;

- upon detection of loss of unscrambled binary 1 (loop initiation signal) from the modem conformance tester, the modem under test shall apply the test loop 2 condition within itself. This condition can be detected at the digital interface of the modem conformance tester as the data signals which it transmits being returned to it. The modem under test shall now indicate to the terminal that the modem is in a test mode (equivalent: turning ON Cct 142).

A.10.2 Deactivation of remote test loop 2

A.10.2.1 Transmission of the deactivation signal(controlling modem under test)

The two modems should be in the configuration obtained when the test described in subclause A.10.1.1 was completed (viz controlling modem under test). Using the means described by the applicant, the modem is commanded to issue a request to de-activate remote test loop 2.

The line signals transmitted by the modem under test are monitored to observe that for not less than 67 ms and not more than 87 ms the level of the transmitted signal at the output of the modem shall fall to below -50 dBm

The modem under test shall now give an indication to the DTE that the modem is no longer in a test mode (equivalent: turning OFF Cct 142).

A.10.2.2 Response to the deactivation signal (controlled modem under test)

The two modems should be in the configuration obtained when the test to subclause A.10.1.2 was completed (viz controlled modem under test).

The signal transmitted by the modem conformance tester is interrupted for 8 ms ± 1 ms. The loop shall remain activated.

The signal transmitted by the modem conformance tester is interrupted for 26 ms ± 1 ms. The modem under test shall release the test loop and restore end to end data transmission within not less than 115 ms and not more than 229 ms, timed from the presentation of the interruption to the modem under test, and give an indication to the DTE that the modem is no longer in a test mode (equivalent: turning OFF Cct 142).

Annex B (normative): Derivation of timings

The following table gives the time, in ascending order, of the various signals mentioned in the requirements and tests of this ETS, together with an explanation of how they were derived when this is not immediately obvious. The first value is the time in ms, followed by in parentheses, the number of bits at 1200 bit/s and, if appropriate, at 600 bit/s.

Table B1

17 ms ± 7 ms 10 ms(12|6) 17 ms(20,4|10,2) 24 ms(28,8|14,4)
77 ms ± 10 ms 67 ms(80,4|40,2) 77 ms(92,4|46,2) 87 ms(104,4|52,2)
154 ms to 231 ms 154 ms(184,8|92,4) 231 ms(277,2|138,6)
172 ms ± 57 ms 115 ms(138|69) 172 ms(206,4|103,2) 229 ms(274,8|137,4) These are the maximum and minimum values of the sum of 17 ms ± 7 ms and 155 ms ± 50 ms.
231 ms to 308 ms 231 ms(277,2|138,6) 308 ms(369,6|184,8)
270 ms ± 40 ms 230 ms(276) 270 ms(324) 310 ms(372)
Minimum of 551 ms 551 ms(661,2) This is derived from the sum of the minimums of 155 ms ± 50 ms (105 ms) and 456 ms ± 10 ms (446 ms) NOTE : the tolerance on the 155 ms is less (i.e. ± 10 ms)in Rec CCITT V.22bis.
765 ms ± 10 ms 755 ms (906) 765 ms(918) 775 ms(930)
1035 ms ± 50 ms 985 ms(1182) 1035 ms (1242) 1085 ms(1302) This is derived from the sum of 270 ms ± 40 ms and 765 ms ± 10 ms
NOTE : Where the modem conformance tester is required to generate signals, the limits placed on the time for which these signals are transmitted are generally the shortest possible, thus making them as difficult as possible to detect. A tolerance of ± 1 ms is specified for the time of such signals to permit integer numbers of bits to be generated.

Annex C (informative): Example proforma for the declaration of modes of operation/use

Declaration of modes of operation/use for 1200 bit/s / 600 bit/s modems based on Recommendation CCITT V.22

The modem submitted for approval shall be capable of operating in the modes indicated in the Tables below.

Proforma for declaring modes of operation/use

Table C1 : transmission at 1200 bit/s

1200 bits per second	
a) Asynchronous with 8 bit characters	
b) Asynchronous with 9 bit characters	
c) Asynchronous with 10 bit characters	
d) Asynchronous with 11 bit characters	
e) Synchronous	

Table C2 : transmission at 600 bit/s

600 bits per second	
a1) Asynchronous with 8 bit characters	
b1) Asynchronous with 9 bit characters	
c1) Asynchronous with 10 bit characters	
d1) Asynchronous with 11 bit characters	
e1) Synchronous	

Table C3 : encoding and decoding of data

Encoding and decoding of data	
Declaration of conformance	

Table C4 : channel selection

Channel Selection	
a) Call Mode	
b) Answer Mode	
Where the answer to both a) and b) is YES The method or methods of mode selection as required by Clause 3.4	
c) Automatic selection by detection of an incoming PSTN call	
d) Selection via the digital interface (equivalent: Cct 126)	
e) Manual selection using means provided on the modem.	

Table C5 : auto-calling and/or answering

Auto calling and/or answering	
a) Auto calling	
b) Auto answering	

Table C6 : test loop

Test Loop	
a) Method of generating initiation signal	
b) Method of de-activating test loop	
c) Means of indicating that the modem is in a test mode	
d) Means of indicating that the modem is no longer in a test mode	

Annex D (informative): Bibliography

The following are references contained within the text of this ETS:

CCITT Recommendation Q.144 (1984): "Specifications of Signalling System N° 5 : Line Signal Receiver".

CCITT Recommendation V.14 (1988): "Transmission of start-stop characters over synchronous bearer channels".

CCITT Recommendation V.22 bis (1988): "2400 bits per second duplex modem using the frequency division technique standardized for use on the General Switched Telephone Network and on point-to-point 2-wire leased telephone-type circuits.

CCITT Recommendation V.24 (1984): "List of definitions for interchange circuits between Data Terminal Equipment and Data Circuit-Terminating Equipment".

CCITT Recommendation V.42 (1988): "Error correcting procedures for DCEs using asychronous-to-synchronous conversion".

CCITT Recommendation V.54 (1984): "Loop test devices for modems".

History

Document history	
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